Electric trucks routing with deterministic and stochastic travel times

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1 Problem statement

We consider a distribution task where a fleet of m electric trucks must deliver feed to a set of n remote silos. Each truck operates over a single shift and is constrained by a limited battery capacity, modeled as a maximum travel-time budget $T_{\max} > 0$. Because this budget is tight, it is typically impossible for the fleet to visit all silos within the same shift, and therefore a selection must be made.

Each silo i is associated with a nonnegative reward (an urgency score) $u_i \geq 0$ and can be visited at most once across the entire fleet, i.e., its reward can be collected only once. Every truck must start from a common depot and return to the same depot before its battery runs out. If the total travel time of a truck exceeds $T_{\max}$, the truck fails to complete its route and all rewards assigned to that truck are lost (all-or-nothing payoff). The overall goal is to design up to m depot-to-depot routes that maximize the total collected reward.

We study two versions of the problem. In the deterministic setting, travel times are given by Euclidean distances between locations. In the stochastic setting, travel times are random variables (modeled as log-normal), capturing uncertainty in rural-road travel times due to factors such as weather and road conditions.

2 Solution

This section presents the complete solution framework adopted in this work. In Subsection 2.1, we introduce the mathematical formulation of the deterministic problem and its stochastic extension, including the exact MILP model (which we will use as a benchmark on small instances). In Subsection 2.2, we describe the algorithmic approach employed to compute solutions in practice, namely a constructive-first heuristic for the deterministic setting and its extension into a simheuristic for the stochastic case. Finally, in Subsection 2.3, we report the computational results obtained for both the deterministic and the stochastic versions of the problem.

2.1 Mathematical model

We model the problem on a complete directed graph.

Let

$$N = \{0, 1, \ldots, n\}$$

be the set of nodes, where 0 denotes the depot and nodes $i = 1, \ldots, n$ represent the silos, with $n \in \mathbb{N} \setminus \{0\}$.

Each node $i \in N$ is associated with planar coordinates $(x_i, y_i) \in \mathbb{R}^2$. Each silo $i \in N \setminus \{0\}$ has a nonnegative reward $u_i \geq 0$, while we set $u_0 := 0$.

The arc set is

$$A := \{(i, j) \in N \times N : i \neq j\},$$

and we denote by

$$a := |A| = 2\binom{n+1}{2} = (n+1)\,n.$$

Let $M = \{1, \ldots, m\}$ be the set of available electric trucks, with $m \in \mathbb{N} \setminus \{0\}$. Each truck has a maximum travel-time budget $T_{\max} > 0$ for the shift.

2.1.1 Decision variables

For each truck $k \in M$ and each arc $(i, j) \in A$, we introduce binary routing variables $\rho_{ij}^k \in \{0, 1\}$:

$$\rho_{ij}^k = \begin{cases} 1, & \text{if truck } k \text{ travels directly from node } i \text{ to node } j, \\ 0, & \text{otherwise.} \end{cases}$$

Implementation note. In the Python implementation, routing variables are denoted by `x[i,j,k]` and correspond to the mathematical variables ρ_{ij}^k introduced above.

2.1.2 Deterministic version

In the deterministic version, travel times are given by Euclidean distances:

$$t_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}, \qquad \forall (i, j) \in A. \tag{1}$$

Since Euclidean distances are symmetric, we have $t_{ij} = t_{ji}$ for all $i, j \in N, i \neq j$.

Objective function. We aim to maximize the total collected reward. For notational convenience, we denote by

$$\rho = \left(\rho_{ij}^k\right)_{(i,j) \in A,\ k \in M} \in \{0, 1\}^{m \times a}$$

the family of binary routing decisions, where $\rho_{ij}^k = 1$ if and only if truck k travels directly from node i to node j. A silo j contributes its reward u_j whenever it is visited, which (under our routing variables) corresponds to selecting an incoming arc into j for some truck. Therefore, the objective function is

$$F_{\text{det}}(\rho) = \sum_{k \in M} \sum_{j \in N \setminus \{0\}} u_j \sum_{\substack{i \in N: \\ i \neq j}} \rho_{ij}^k. \tag{2}$$

Constraints. We now list the constraints of the deterministic model.

For each truck $k \in M$, the total travel time along its selected arcs must not exceed $T_{\max}$:

$$\sum_{(i,j)\in A} t_{ij}\, \rho_{ij}^k \;\le\; T_{\max}, \qquad \forall k \in M. \tag{3}$$

Each silo can be assigned to at most one truck (and visited at most once):

$$\sum_{k\in M} \sum_{\substack{i\in N:\\ i\neq j}} \rho_{ij}^k \;\le\; 1, \qquad \forall j \in N \setminus \{0\}. \tag{4}$$

Each truck leaves the depot exactly once and returns to the depot exactly once:

$$\sum_{j\in N\setminus\{0\}} \rho_{0j}^k = 1, \qquad \sum_{i\in N\setminus\{0\}} \rho_{i0}^k = 1, \qquad \forall k \in M. \tag{5}$$

For each silo $h \in N \setminus \{0\}$ and each truck $k \in M$, if the truck enters h then it must also leave h (and vice versa):

$$\sum_{\substack{i\in N:\\ i\neq h}} \rho_{ih}^k \;=\; \sum_{\substack{j\in N:\\ j\neq h}} \rho_{hj}^k, \qquad \forall h \in N \setminus \{0\},\ \forall k \in M. \tag{6}$$

Feasible set. We define the deterministic feasible set as

$$\mathcal{X}_{\text{det}} := \left\{ \rho \in \{0,1\}^{m\times a} \;:\; \rho \text{ satisfies (3)–(6)} \right\}.$$

Therefore, the deterministic model can be written compactly as

$$\max_{\rho\in\mathcal{X}_{\text{det}}} \; F_{\text{det}}(\rho), \tag{7}$$

where $F(\rho)$ is defined in (2).

Exact MILP formulation. To turn the deterministic model introduced above into an exact arc-based mixed-integer linear program, we add the missing constraints needed to (i) allow unused trucks and (ii) eliminate disconnected subtours.

To avoid forcing every truck to be used (which, together with (4), may even make the model infeasible when $m > n$), we introduce binary activation variables

$$\delta_k \in \{0,1\} \qquad \forall k \in M,$$

and replace (5) with

$$\sum_{j\in N\setminus\{0\}} \rho_{0j}^k = \delta_k, \qquad \sum_{i\in N\setminus\{0\}} \rho_{i0}^k = \delta_k, \qquad \forall k \in M.$$

It is then natural to link the time budget to the activation decision, by imposing

$$\sum_{(i,j)\in A} t_{ij}\, \rho_{ij}^k \le T_{\max}\, \delta_k, \qquad \forall k \in M,$$

so that when $\delta_k = 0$ the truck is inactive and no travel is allowed, while when $\delta_k = 1$ the usual time budget $T_{\max}$ applies.

Implementation note. In the Python implementation, the truck-activation variables are denoted by `yk[k]` in the code and correspond to the activation indicators δ_k used in this report.

The constraints (5)–(6) enforce flow conservation at each silo for each truck and ensure that each active truck leaves and returns to the depot exactly once. However, in a pure arc-based MILP formulation these constraints do not exclude the possibility of disconnected subtours, i.e., cycles on a subset of silos that are not connected to the depot but still satisfy flow conservation. In that case, the model could artificially collect rewards from nodes in such cycles without actually building a valid depot-to-depot route.

A standard way to exclude subtours is to add MTZ-type ordering variables. For each truck k and each silo $i \in N \setminus \{0\}$, let w_i^k denote the visit order (continuous variables are sufficient, but they can also be declared integer):

$$w_i^k \geq 0 \qquad \forall i \in N \setminus \{0\},\ \forall k \in M.$$

Introduce the binary visit indicators

$$v_i^k := \sum_{j \in N} \rho_{ij}^k \in \{0,1\} \qquad \forall i \in N \setminus \{0\},\ \forall k \in M,$$

and link them to activation by $v_i^k \leq \delta_k$. Then impose the MTZ bounds and inequalities

$$v_i^k \leq w_i^k \leq n\, v_i^k, \qquad \forall i \in N \setminus \{0\},\ \forall k \in M,$$

$$w_i^k - w_j^k + n\, \rho_{ij}^k \leq n - 1, \qquad \forall i \neq j \in N \setminus \{0\},\ \forall k \in M.$$

Together with (5)–(6), these constraints eliminate disconnected subtours and yield an exact MILP formulation.

In this work, solutions are primarily generated by route-based heuristics that explicitly construct, for each truck, a single depot-to-depot route as an ordered sequence of visited silos while maintaining feasibility with respect to the time budget $T_{\max}$. Therefore, disconnected subtours cannot be produced by the algorithmic framework, and explicit subtour-elimination constraints are not required in the heuristic pipeline. The exact MILP is used only as a post-processing benchmark on a small instance (50 silos plus depot): the heuristic solution is provided to the solver as a warm start, and we check whether the solver can improve it. As reported in the Results subsection 2.3, no improvement is obtained on that instance.

2.1.3 Stochastic version

In the stochastic setting, travel times are random variables. For each arc $(i,j) \in A$, let

$$T_{ij} \sim \mathrm{LogNormal}(\mu_{ij}, \sigma_{ij}^2).$$

We calibrate the lognormal law so that the mean equals the deterministic travel time introduced in (3), and the variance is proportional to it:

$$\mathbb{E}[T_{ij}] = t_{ij}, \qquad \mathrm{Var}(T_{ij}) = c\, t_{ij},$$

where $c \geq 0$ is a variability parameter (deterministic instances correspond to $c = 0$; in the experiments $c = 0.05$). Recalling that, for $T \sim \mathrm{LogNormal}(\mu, \sigma^2)$, one has $\mathbb{E}[T] = \exp(\mu + \frac{1}{2}\sigma^2)$ and $\mathrm{Var}(T) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$, the above moment conditions are enforced by the closed-form choice

$$\sigma_{ij}^2 = \ln\left(1 + \frac{c}{t_{ij}}\right), \qquad \mu_{ij} = \ln(t_{ij}) - \frac{1}{2}\sigma_{ij}^2, \qquad \forall (i,j) \in A.$$

Since t_{ij} is the Euclidean distance, we have $t_{ij} = t_{ji}$ whenever both (i,j) and (j,i) belong to A. With the calibration $\mathbb{E}[T_{ij}] = t_{ij}$ and $\mathrm{Var}(T_{ij}) = c\, t_{ij}$, this implies $T_{ij} \overset{d}{=} T_{ji}$.

The family $\{T_{ij}\}_{(i,j)\in A}$ is assumed mutually independent, meaning that in each Monte Carlo run travel times on different arcs are sampled independently.

All-or-nothing reward under battery failure. Given a solution (i.e., a set of selected arcs ρ^k), the random travel time of truck k is

$$S_k(\rho, T) = \sum_{(i,j)\in A} T_{ij}\, \rho^k_{ij}.$$

If $S_k > T_{\max}$, truck k fails to return to the depot and all rewards collected by that truck are lost. Let the (deterministic) reward collected by truck k be

$$U_k(\rho) = \sum_{j\in N\setminus\{0\}} u_j \sum_{\substack{i\in N:\\ i\neq j}} \rho^k_{ij}.$$

The stochastic collected reward of truck k is then

$$\widetilde{U}_k(\rho, T) = U_k(\rho)\, \mathbf{1}\{S_k(\rho, T) \le T_{\max}\}.$$

Objective function. The stochastic objective function to be maximized is the expected total reward:

$$F_{\text{stoc}}(\rho) \;=\; \sum_{k\in M} \mathbb{E}\big[\widetilde{U}_k(\rho, T)\big] \;=\; \sum_{k\in M} U_k(\rho)\, \mathbb{P}\big(S_k(\rho, T) \le T_{\max}\big),$$

where we used that $U_k(\rho)$ is deterministic once the routing decision ρ is fixed.

Since $S_k(\rho, T)$ is a sum of lognormal random variables, the probability term typically has no closed form and will be estimated via Monte Carlo simulation within the algorithmic framework described next.

Feasible set and compact formulation. Routing decisions are required to satisfy the same feasibility constraints as in the deterministic model, except for the deterministic time-budget constraint (3), which is now handled stochastically through the probability of completing each route within $T_{\max}$. We define

$$\mathcal{X}_{\text{stoc}} := \big\{\rho \in \{0,1\}^{m\times a} \;:\; \rho \text{ satisfies (5)–(6)}\big\},$$

where $m = |M|$ and $a = |A|$.

Then the stochastic problem can be written compactly as

$$\max_{\rho\in\mathcal{X}_{\text{stoc}}} F_{\text{stoc}}(\rho).$$

2.2 Algorithms

This section describes the algorithmic framework adopted to compute solutions for both the deterministic and the stochastic versions of the problem. We follow a constructive-first approach: a fast heuristic builds, for each truck, a feasible depot-to-depot route as an ordered sequence of visited silos, and the resulting solution is then refined by local improvement moves (e.g., intra-route 2-opt, reinsertion, and swap-type moves). The stochastic extension is presented next: solutions are still constructed as feasible routes, but they are evaluated under random travel times through Monte Carlo simulation.

2.2.1 Deterministic heuristic

In the deterministic setting, we adopt a savings-based construction heuristic inspired by the approach described in Section 4.1 of Panadero et al. [1], adapted to the depot-to-depot case. The construction is followed by an intra-route 2-opt improvement. We then apply a greedy reinsertion procedure to exploit any remaining slack in the time budgets by inserting additional unvisited silos. Finally, we apply a replacement step based on swaps of the form visited $\leftrightarrow$ unvisited, which can improve the total reward even when no further feasible insertions exist.

Route representation and length. A route is represented as an ordered sequence $(0, i_1, \ldots, i_p, 0)$, with $i_\ell \in N \setminus \{0\}$ all distinct. Its deterministic travel time is

$$L(0, i_1, \ldots, i_p, 0) := \sum_{\ell=0}^{p} t_{i_\ell i_{\ell+1}}, \qquad i_0 = i_{p+1} = 0,$$

and it is feasible if $L(\cdot) \leq T_{\max}$.

Savings-based construction. After discarding any silo i such that $t_{0i} + t_{i0} > T_{\max}$, we initialize a dummy solution with one route per remaining silo, namely $(0, i, 0)$. For any ordered pair of distinct silos $i \neq j$, we define the time-based saving

$$s_{ij} := t_{i0} + t_{0j} - t_{ij},$$

and the combined saving score

$$s'_{ij} := \alpha\, s_{ij} + (1 - \alpha)\,(u_i + u_j), \qquad \alpha \in (0, 1).$$

In practice, the parameter α can be tuned through a small grid search, and the value yielding the highest deterministic reward (ties broken by total travel time) is retained.

We sort all ordered pairs (i, j) in non-increasing order of s'_{ij} and scan the list to attempt route merges. A merge connecting i to j is admissible only if i is currently the last silo of its route and j is currently the first silo of its route, the two routes are distinct, and the merged route remains feasible. In the depot-to-depot setting, if a route R_i ends with i and a route R_j starts with j, the merged route length can be computed as

$$L(R_i \oplus_{i \to j} R_j) \;=\; L(R_i) + L(R_j) - t_{i0} - t_{0j} + t_{ij},$$

which must be $\leq T_{\max}$. After no further admissible merges exist, we rank the constructed routes by collected reward and keep the best m routes (discarding the others).

Intra-route 2-opt. We apply an intra-route 2-opt local search to each selected route, accepting only moves that strictly decrease the route travel time (and hence preserve feasibility). This step improves the within-route ordering without changing the set of visited silos.

Greedy reinsertion of unvisited silos. Let $\mathcal{R}$ be the current set of routes, and let U be the set of unvisited silos. For a route $r = (0, i_1, \ldots, i_p, 0)$ and a candidate silo $j \in U$, inserting j between consecutive nodes $(i_\ell, i_{\ell+1})$ yields the incremental travel time

$$\Delta t(j; r, \ell) := t_{i_\ell j} + t_{j i_{\ell+1}} - t_{i_\ell i_{\ell+1}}, \qquad \ell = 0, \ldots, p,$$

where $i_0 = i_{p+1} = 0$. Among all admissible insertion moves (those keeping the updated route length $\leq T_{\max}$), we select the move maximizing a benefit–cost score, e.g.

$$\text{score}(j; r, \ell) = \frac{u_j}{\max(\Delta t(j; r, \ell), \varepsilon)},$$

for a small $\varepsilon > 0$, treating moves with $\Delta t(j; r, \ell) \leq 0$ as dominating options. We repeat until no admissible insertion exists.

Replacement swaps (visited $\leftrightarrow$ unvisited). As a final improvement step, we consider replacement moves that remove one visited silo h from a route and insert one unvisited silo j into the resulting route in the best feasible position (i.e., minimizing the associated insertion cost). Such a move is accepted only if it strictly improves the total reward (e.g. $u_j > u_h$) and preserves feasibility. We iterate replacement swaps until no improving move is found.

Algorithm 1: Deterministic heuristic: savings construction + 2-opt + greedy reinsertion + replacement swaps

Require: Coordinates (x_i, y_i), rewards u_i for $i \in N \setminus \{0\}$, fleet size m, budget $T_{\max}$, parameter $\alpha \in (0,1)$
Ensure: A set of at most m feasible depot-to-depot routes

Preprocessing: remove any silo i with $t_{0i} + t_{i0} > T_{\max}$
Initialize dummy routes $\mathcal{R} \leftarrow \{(0,i,0) : i \in N \setminus \{0\}\}$
Compute and sort the savings list $\mathcal{L}$ of ordered pairs (i,j), $i \neq j$, by decreasing s'_{ij}
Savings-based merging
for all $(i,j) \in \mathcal{L}$ **do**
  Let R_i be the current route ending at i and R_j the current route starting at j
  if $R_i \neq R_j$ **and** $L(R_i) + L(R_j) - t_{i0} - t_{0j} + t_{ij} \leq T_{\max}$ **then**
    Merge R_i and R_j by connecting $i \to j$; update $\mathcal{R}$
  end if
end for
Keep the best m routes in $\mathcal{R}$ by collected reward (ties by shorter travel time)
Intra-route refinement
for all $r \in \mathcal{R}$ **do**
  Apply intra-route 2-opt to r until no improving move exists
end for
Greedy reinsertion of unvisited silos
$V \leftarrow$ set of silos visited by routes in $\mathcal{R}$; $U \leftarrow (N \setminus \{0\}) \setminus V$
while there exists an admissible insertion of some $j \in U$ into some $r \in \mathcal{R}$ **do**
  Choose the admissible insertion move (j^*, r^*, ℓ^*) maximizing score $= u_j / \max(\Delta t(j; r, \ell), \varepsilon)$
  Insert j^* into r^* at position ℓ^*; update $V \leftarrow V \cup \{j^*\}$ and $U \leftarrow U \setminus \{j^*\}$
end while
Replacement swaps (visited $\leftrightarrow$ unvisited)
while there exists an improving replacement move **do**
  Find a feasible move that removes a visited silo h and inserts an unvisited silo j with $u_j > u_h$
  Among all such moves, choose one maximizing $u_j - u_h$ (ties by smaller resulting route length)
  Apply the replacement; update V and U
end while
return $\mathcal{R}$

2.2.2 Stochastic Simheuristic

To handle uncertainty, we adopt a simheuristic approach, i.e., we combine a randomized heuristic search with Monte Carlo simulation in order to estimate the stochastic objective and select robust solutions.

Randomized multi-start construction (Top-L_{top} selection). We generate a pool of candidate solutions by running a randomized variant of the deterministic heuristic. In particular, during the greedy reinsertion phase and during replacement swaps, we build a list of admissible moves and rank them according to their deterministic merit (e.g., reward gain per additional deterministic travel time, while preserving feasibility with respect to $T_{\max}$ under t_{ij}). Instead of always selecting the best move, we consider the Top-L_{top} moves in the ranking and choose one at random. Repeating this procedure for K starts produces K candidate solutions $\rho^{(1)}, \ldots, \rho^{(K)}$.

The savings-based construction and the intra-route 2-opt phases remain deterministic, while randomization is introduced only during the reinsertion and replacement steps.

Monte Carlo evaluation of the stochastic objective. For a fixed candidate solution $\rho^{(q)}$, we estimate the stochastic objective

$$F_{\text{stoc}}(\rho^{(q)}) = \sum_{k \in M} \mathbb{E}\big[\widetilde{U}_k(\rho^{(q)}, T)\big]$$

via Monte Carlo simulation. For $s = 1, \ldots, N$, we sample an independent realization $T^{(s)} = \{T_{ij}^{(s)}\}_{(i,j)\in A}$, compute for each truck

$$S_k^{(s)}(\rho^{(q)}) = \sum_{(i,j)\in A} T_{ij}^{(s)}\, \rho_{ij}^{k,(q)}, \qquad \widetilde{U}_k^{(s)}(\rho^{(q)}) = U_k(\rho^{(q)})\, \mathbf{1}\{S_k^{(s)}(\rho^{(q)}) \le T_{\max}\},$$

and obtain the sample total reward $\widetilde{U}^{(s)}(\rho^{(q)}) = \sum_{k\in M} \widetilde{U}_k^{(s)}(\rho^{(q)})$. The expected total reward is then estimated as

$$\widehat{F}_{\text{stoc}}(\rho^{(q)}) = \frac{1}{N}\sum_{s=1}^{N} \widetilde{U}^{(s)}(\rho^{(q)}).$$

Reliability metric and selection rule. Along with the expected reward, we compute a reliability indicator measuring the route success rate. In particular, for each truck we estimate the probability of completing the route within the time budget,

$$\widehat{p}_k(\rho^{(q)}) = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}\{S_k^{(s)}(\rho^{(q)}) \le T_{\max}\}.$$

and define the solution reliability as the average success rate across trucks,

$$\widehat{R}(\rho^{(q)}) = \frac{1}{|M|}\sum_{k\in M} \widehat{p}_k(\rho^{(q)}) \in [0,1].$$

Implementation note. In our implementation, unused trucks are represented as empty depot-to-depot routes (i.e., $0 \to 0$ with zero travel time). Therefore, they always satisfy the time budget and contribute $\widehat{p}_k = 1$ to the average reliability $\widehat{R}$.

Given a target reliability level $\beta \in (0,1)$, we select the final solution by solving the empirical rule

$$\max\ \widehat{F}_{\text{stoc}}(\rho^{(q)}) \qquad \text{s.t.} \qquad \widehat{R}(\rho^{(q)}) \ge \beta, \qquad q \in \{1, \ldots, K\}.$$

If no candidate satisfies $\widehat{R}(\rho^{(q)}) \ge \beta$, we return the most reliable candidate (breaking ties by expected reward).

Algorithm 2: Stochastic simheuristic: randomized multi-start + Monte Carlo evaluation + reliability-based selection

Require: Instance data, fleet size m (with $M = \{1, \ldots, m\}$), time budget $T_{\max}$, deterministic heuristic parameter $\alpha \in (0,1)$, number of starts $K \in \mathbb{N}$, Top-L parameter $L_{\text{top}} \in \mathbb{N}$, Monte Carlo sample size $N \in \mathbb{N}$, reliability threshold $\beta \in (0,1)$

Ensure: A feasible depot-to-depot routing solution

Candidate generation (randomized multi-start)

for $q = 1, \ldots, K$ **do**
  Run Algorithm 1 in randomized mode to obtain a feasible candidate $\rho^{(q)}$:
   during the reinsertion phase and during replacement swaps, select one move uniformly at random among the Top-L_{top} admissible moves ranked by the deterministic score.
  Precompute $U_k(\rho^{(q)})$ for all $k \in M$
end for

Monte Carlo evaluation

for $q = 1, \ldots, K$ **do**
  $\widehat{F}_{\text{stoc}}(\rho^{(q)}) \leftarrow 0$
  Initialize counters $c_k \leftarrow 0$ for all $k \in M$
  for $s = 1, \ldots, N$ **do**
    Sample travel times $T^{(s)} = \{T_{ij}^{(s)}\}_{(i,j)\in A}$

$\widetilde{U}^{(s)} \leftarrow 0$
for all $k \in M$ **do**
$S_k^{(s)} \leftarrow \sum_{(i,j)\in A} T_{ij}^{(s)}\, \rho_{ij}^{k,(q)}$
if $S_k^{(s)} \le T_{\max}$ **then**
$\widetilde{U}^{(s)} \leftarrow \widetilde{U}^{(s)} + U_k(\rho^{(q)})$
$c_k \leftarrow c_k + 1$
end if
end for
$\widehat{F}_{\text{stoc}}(\rho^{(q)}) \leftarrow \widehat{F}_{\text{stoc}}(\rho^{(q)}) + \widetilde{U}^{(s)}$
end for
$\widehat{F}_{\text{stoc}}(\rho^{(q)}) \leftarrow \widehat{F}_{\text{stoc}}(\rho^{(q)})/N$
$\widehat{p}_k(\rho^{(q)}) \leftarrow c_k/N \quad \forall k \in M$
$\widehat{R}(\rho^{(q)}) \leftarrow \frac{1}{|M|}\sum_{k\in M} \widehat{p}_k(\rho^{(q)})$
end for
Selection
$\mathcal{Q} \leftarrow \{q \in \{1,\dots,K\} : \widehat{R}(\rho^{(q)}) \ge \beta\}$
if $\mathcal{Q} \neq \emptyset$ **then**
Choose $q^\star \in \arg\max_{q\in\mathcal{Q}} \widehat{F}_{\text{stoc}}(\rho^{(q)})$
else
Choose $q^\star \in \arg\max_{q\in\{1,\dots,K\}} \widehat{R}(\rho^{(q)})$
end if
return $\rho^{(q^\star)}$

2.3 Results

We report results on nine instances with $n = 98$ silos and 1 depot and $m \in \{2, 3, 4\}$ trucks. For each value of m, we consider three time-budget regimes (tight/medium/loose) by varying $T_{\max}$. Table 1 compares the deterministic baseline (*Heuristic + MC*) - a solution constructed using deterministic travel times and then evaluated under stochastic travel times - against the *Simheuristic*, which embeds Monte Carlo estimates within the search.

For each method, we report the deterministic *Reward* (sum of rewards of visited silos), the Monte Carlo estimate of the expected reward $\widehat{\mathbb{E}}[\text{Reward}]$, the estimated *Reliability* $\widehat{R}$ (average route success rate across trucks), and the computing time. The column *Impr%* (percent improvement) is computed from the expected reward estimates as

$$\text{Impr\%} \;=\; 100 \cdot \frac{\widehat{\mathbb{E}}[\text{Reward}]_{\text{Sim}} - \widehat{\mathbb{E}}[\text{Reward}]_{\text{Base}}}{\widehat{\mathbb{E}}[\text{Reward}]_{\text{Base}}}\,,$$

where "Base" denotes the deterministic baseline evaluated under uncertainty and "Sim" denotes the simheuristic solution.

Impact of the time budget. Across all values of m, the simheuristic delivers its most pronounced gains in the tight-to-medium regimes, where route failures are more likely and the all-or-nothing reward evaluation makes robustness particularly valuable; nevertheless, sizable improvements can also persist in the loose regime. For $m = 2$, the simheuristic substantially improves $\widehat{\mathbb{E}}[\text{Reward}]$ and increases reliability in all regimes: on the tight instance `p4.2.b` ($T_{\max} = 30$) and the medium instance `p4.2.h` ($T_{\max} = 60$), reliability rises from 0.706 and 0.5525 to about 0.918 and 0.8455, respectively, with improvements in expected reward ranging from about 25% to 39%. Even in the loose regime (`p4.2.t`, $T_{\max} = 120$), the simheuristic pushes reliability above 0.93 while increasing the expected reward by about 32%.

A similar pattern appears for $m = 3$. On `p4.3.b` ($T_{\max} = 20$), the simheuristic attains a large improvement in $\widehat{\mathbb{E}}[\text{Reward}]$ together with a substantial reliability gain (from 0.6453 to 0.849), despite only minor changes in deterministic reward and visited silos. On `p4.3.k` ($T_{\max} = 50$), the baseline is already relatively reliable, hence the additional benefit is more modest, yet still positive, and reliability exceeds 0.93. On `p4.3.t` ($T_{\max} = 80$), the improvement remains positive and reliability increases to about 0.913, suggesting that stochastic guidance can still help identify route structures that remain robust even under relaxed budgets.

Table 1: Stochastic results summary: Monte Carlo with $N = 1000$, $K = 120$, $Top - L = 15$, $c = 0.05$ and $\beta = 0.9$.

Instance	n	m	$T_{\max}$	Algorithm	#Visited	Reward	$\mathbb{E}$[Reward]	Reliability	%Impr.	Time (s)
p4.2.b	98	2	30.00	Heuristic + MC	20	358.00	240.58	0.7060		0.025
p4.2.b	98	2	30.00	Simheuristic	18	326.00	300.40	0.9180	24.9	12.806
p4.2.h	98	2	60.00	Heuristic + MC	43	697.00	385.17	0.5525		0.083
p4.2.h	98	2	60.00	Simheuristic	37	648.00	536.09	0.8455	39.2	49.514
p4.2.t	98	2	120.00	Heuristic + MC	92	1207.00	843.42	0.6815		0.099
p4.2.t	98	2	120.00	Simheuristic	91	1201.00	1114.37	0.9300	32.1	90.306
p4.3.b	98	3	20.00	Heuristic + MC	18	317.00	197.29	0.6453		0.007
p4.3.b	98	3	20.00	Simheuristic	18	316.00	271.38	0.8490	37.6	8.746
p4.3.k	99	3	50.00	Heuristic + MC	43	716.00	631.58	0.8637		0.081
p4.3.k	99	3	50.00	Simheuristic	41	707.00	669.78	0.9383	6.0	78.133
p4.3.t	98	3	80.00	Heuristic + MC	87	1203.00	879.79	0.7357		0.302
p4.3.t	98	3	80.00	Simheuristic	68	1124.00	1033.55	0.9127	17.5	131.984
p4.4.d	98	4	20.00	Heuristic + MC	21	359.00	219.42	0.6158		0.007
p4.4.d	98	4	20.00	Simheuristic	21	356.00	282.81	0.7620	28.9	7.832
p4.4.l	98	4	40.00	Heuristic + MC	40	707.00	429.22	0.6420		0.059
p4.4.l	98	4	40.00	Simheuristic	39	669.00	583.87	0.8785	36.0	28.049
p4.4.t	98	4	60.00	Heuristic + MC	77	1165.00	710.96	0.6185		0.074
p4.4.t	98	4	60.00	Simheuristic	70	1115.00	920.16	0.8087	29.4	59.984

For $m = 4$, the simheuristic again produces consistent gains. In the medium and loose instances (`p4.4.l`, $T_{\max} = 40$ and `p4.4.t`, $T_{\max} = 60$), the expected reward increases substantially (about 36% and 29%, respectively) while reliability improves from the baseline level to roughly 0.879 and 0.809. In the tighter instance `p4.4.d` ($T_{\max} = 20$), the simheuristic still delivers a meaningful improvement (about 29%), reflecting the difficulty of constructing highly reliable routes under stringent time limits when four trucks are available.

Solution structure and trade-offs. Overall, the simheuristic achieves higher expected reward primarily by increasing route feasibility under uncertainty. This "robustification" often manifests as a reduction in the number of visited silos and/or a reshaping of route compositions, which increases slack and thus the probability of completing each route within $T_{\max}$. Importantly, the deterministic *Reward* can decrease while $\widehat{\mathbb{E}}$[Reward] increases, highlighting that under all-or-nothing evaluation the key driver of performance is reliability rather than the deterministic proxy.

Computational effort. As expected, the simheuristic is computationally more demanding than the baseline heuristic since it performs repeated Monte Carlo evaluations during the search. Nevertheless, for the tested instances the additional runtime remains within a practical range and is accompanied by consistent improvements in expected reward and reliability. Overall, these results confirm that explicitly incorporating stochastic evaluation within the search can substantially improve solution quality under uncertainty, especially in tight-to-medium time-budget regimes.

Table 2: Route paths for instance `p4.2.b`.

Algorithm	Truck	#Visited	Route
Heuristic	1	11	0 → 96 → 50 → 3 → 86 → 92 → 47 → 23 → 7 → 14 → 67 → 9 → 0
Heuristic	2	9	0 → 57 → 81 → 29 → 42 → 53 → 77 → 72 → 73 → 30 → 0
Simheuristic	1	11	0 → 14 → 34 → 7 → 23 → 47 → 50 → 3 → 92 → 86 → 41 → 96 → 0
Simheuristic	2	7	0 → 30 → 73 → 16 → 54 → 58 → 40 → 64 → 0

Table 3: Route paths for instance `p4.2.h`.

Algorithm	Truck	#Visited	Route
Heuristic	1	19	0 → 30 → 73 → 16 → 64 → 40 → 58 → 54 → 72 → 77 → 27 → 83 → 95 → 51 → 59 → 42 → 53 → 29 → 9 → 57 → 0
Heuristic	2	24	0 → 96 → 50 → 3 → 47 → 23 → 7 → 14 → 67 → 10 → 69 → 80 → 22 → 88 → 71 → 12 → 46 → 25 → 91 → 61 → 28 → 38 → 20 → 79 → 81 → 0
Simheuristic	1	21	0 → 23 → 7 → 14 → 67 → 9 → 79 → 81 → 29 → 42 → 59 → 51 → 95 → 83 → 27 → 77 → 72 → 54 → 64 → 16 → 73 → 30 → 0
Simheuristic	2	16	0 → 34 → 87 → 84 → 63 → 70 → 33 → 11 → 18 → 94 → 92 → 86 → 41 → 96 → 50 → 3 → 47 → 0

Table 4: Route paths for instance `p4.2.t`.

Algorithm	Truck	#Visited	Route
Heuristic	1	49	0 → 14 → 67 → 9 → 79 → 81 → 29 → 53 → 42 → 72 → 77 → 27 → 95 → 83 → 56 → 93 → 51 → 59 → 20 → 32 → 26 → 38 → 28 → 61 → 37 → 91 → 25 → 46 → 5 → 13 → 1 → 75 → 90 → 15 → 2 → 65 → 48 → 17 → 12 → 71 → 88 → 22 → 80 → 69 → 10 → 85 → 52 → 34 → 7 → 23 → 0
Heuristic	2	43	0 → 30 → 73 → 57 → 43 → 87 → 84 → 60 → 98 → 6 → 97 → 55 → 4 → 39 → 62 → 74 → 45 → 19 → 31 → 8 → 68 → 36 → 78 → 24 → 82 → 76 → 89 → 49 → 44 → 35 → 70 → 33 → 63 → 11 → 18 → 94 → 92 → 47 → 3 → 50 → 86 → 41 → 66 → 96 → 0
Simheuristic	1	50	0 → 14 → 67 → 9 → 79 → 81 → 29 → 53 → 42 → 72 → 77 → 27 → 95 → 83 → 93 → 51 → 59 → 20 → 32 → 26 → 38 → 28 → 61 → 37 → 91 → 25 → 46 → 5 → 13 → 1 → 75 → 90 → 15 → 2 → 65 → 48 → 17 → 12 → 71 → 88 → 22 → 80 → 69 → 10 → 85 → 52 → 34 → 7 → 23 → 47 → 96 → 0
Simheuristic	2	41	0 → 30 → 73 → 57 → 43 → 87 → 84 → 60 → 98 → 6 → 97 → 55 → 4 → 39 → 62 → 74 → 45 → 19 → 31 → 8 → 68 → 36 → 78 → 24 → 82 → 76 → 89 → 49 → 44 → 35 → 70 → 33 → 63 → 11 → 18 → 94 → 92 → 86 → 3 → 50 → 41 → 66 → 0

Table 5: Route paths for instance `p4.3.b`.

Algorithm	Truck	#Visited	Route
Heuristic	1	7	0 → 96 → 47 → 50 → 3 → 92 → 86 → 41 → 0
Heuristic	2	6	0 → 23 → 7 → 34 → 14 → 67 → 9 → 0
Heuristic	3	5	0 → 79 → 29 → 81 → 57 → 30 → 0
Simheuristic	1	7	0 → 96 → 41 → 86 → 92 → 3 → 50 → 47 → 0
Simheuristic	2	5	0 → 9 → 81 → 29 → 79 → 67 → 0
Simheuristic	3	6	0 → 30 → 73 → 57 → 14 → 7 → 23 → 0

Table 6: Route paths for instance `p4.3.k`.

Algorithm	Truck	#Visited	Route
Heuristic	1	16	0 → 9 → 67 → 79 → 81 → 29 → 53 → 42 → 59 → 51 → 95 → 83 → 27 → 77 → 72 → 57 → 30 → 0
Heuristic	2	14	0 → 14 → 34 → 43 → 7 → 23 → 96 → 41 → 66 → 64 → 40 → 58 → 54 → 16 → 73 → 0
Heuristic	3	13	0 → 47 → 50 → 3 → 86 → 92 → 11 → 63 → 33 → 70 → 84 → 87 → 18 → 94 → 0
Simheuristic	1	17	0 → 14 → 67 → 79 → 81 → 29 → 53 → 42 → 59 → 51 → 95 → 83 → 27 → 77 → 72 → 57 → 73 → 30 → 0
Simheuristic	2	11	0 → 7 → 47 → 3 → 50 → 96 → 64 → 40 → 58 → 54 → 16 → 9 → 0
Simheuristic	3	13	0 → 23 → 34 → 87 → 84 → 60 → 70 → 63 → 18 → 94 → 92 → 86 → 41 → 66 → 0

Table 7: Route paths for instance `p4.3.t`.

Algorithm	Truck	#Visited	Route
Heuristic	1	23	0 → 57 → 81 → 29 → 59 → 83 → 95 → 27 → 51 → 42 → 53 → 77 → 72 → 54 → 58 → 40 → 64 → 16 → 73 → 30 → 66 → 41 → 86 → 92 → 0
Heuristic	2	31	0 → 14 → 67 → 10 → 69 → 80 → 22 → 88 → 71 → 12 → 17 → 48 → 2 → 15 → 90 → 75 → 1 → 13 → 5 → 46 → 25 → 91 → 93 → 37 → 61 → 28 → 38 → 20 → 26 → 32 → 79 → 9 → 0
Heuristic	3	33	0 → 96 → 50 → 3 → 47 → 23 → 18 → 94 → 87 → 84 → 60 → 21 → 98 → 6 → 76 → 82 → 24 → 78 → 36 → 68 → 8 → 31 → 19 → 45 → 74 → 55 → 62 → 39 → 4 → 85 → 52 → 34 → 43 → 7 → 0
Simheuristic	1	24	0 → 67 → 10 → 80 → 32 → 26 → 20 → 38 → 28 → 79 → 29 → 81 → 42 → 59 → 51 → 95 → 83 → 27 → 77 → 54 → 58 → 40 → 64 → 16 → 30 → 0
Simheuristic	2	20	0 → 73 → 72 → 57 → 9 → 14 → 7 → 87 → 84 → 70 → 33 → 63 → 18 → 94 → 92 → 86 → 3 → 50 → 41 → 96 → 23 → 0
Simheuristic	3	24	0 → 34 → 76 → 82 → 24 → 78 → 74 → 45 → 8 → 31 → 19 → 48 → 17 → 75 → 15 → 90 → 1 → 13 → 5 → 12 → 88 → 22 → 69 → 85 → 52 → 0

Table 8: Route paths for instance `p4.4.d`.

Algorithm	Truck	#Visited	Route
Heuristic	1	7	0 → 96 → 47 → 50 → 3 → 92 → 86 → 41 → 0
Heuristic	2	6	0 → 23 → 7 → 34 → 14 → 67 → 9 → 0
Heuristic	3	5	0 → 79 → 29 → 81 → 57 → 30 → 0
Heuristic	4	3	0 → 87 → 18 → 94 → 0
Simheuristic	1	7	0 → 96 → 41 → 86 → 92 → 3 → 50 → 47 → 0
Simheuristic	2	5	0 → 57 → 81 → 29 → 9 → 67 → 0
Simheuristic	3	5	0 → 30 → 73 → 14 → 34 → 7 → 0
Simheuristic	4	4	0 → 87 → 94 → 18 → 23 → 0

Table 9: Route paths for instance `p4.4.l`.

Algorithm	Truck	#Visited	Route
Heuristic	1	12	$0 \to 96 \to 50 \to 3 \to 92 \to 47 \to 23 \to 7 \to 14 \to 67 \to 10 \to 79 \to 9 \to 0$
Heuristic	2	10	$0 \to 30 \to 72 \to 77 \to 27 \to 83 \to 95 \to 42 \to 29 \to 81 \to 57 \to 0$
Heuristic	3	8	$0 \to 73 \to 16 \to 54 \to 58 \to 40 \to 64 \to 41 \to 86 \to 0$
Heuristic	4	10	$0 \to 34 \to 43 \to 87 \to 84 \to 60 \to 21 \to 70 \to 63 \to 18 \to 94 \to 0$
Simheuristic	1	13	$0 \to 14 \to 67 \to 9 \to 79 \to 29 \to 81 \to 57 \to 30 \to 86 \to 92 \to 3 \to 50 \to 96 \to 0$
Simheuristic	2	9	$0 \to 73 \to 72 \to 77 \to 27 \to 83 \to 95 \to 59 \to 42 \to 53 \to 0$
Simheuristic	3	7	$0 \to 16 \to 54 \to 58 \to 40 \to 64 \to 41 \to 66 \to 0$
Simheuristic	4	10	$0 \to 7 \to 34 \to 98 \to 60 \to 84 \to 87 \to 94 \to 18 \to 47 \to 23 \to 0$

Table 10: Route paths for instance `p4.4.t`.

Algorithm	Truck	#Visited	Route
Heuristic	1	21	$0 \to 47 \to 23 \to 7 \to 14 \to 9 \to 79 \to 29 \to 53 \to 42 \to 59 \to 51 \to 95 \to 83 \to 27 \to 77 \to 72 \to 54 \to 64 \to 16 \to 73 \to 30 \to 0$
Heuristic	2	18	$0 \to 43 \to 87 \to 84 \to 60 \to 21 \to 63 \to 70 \to 33 \to 11 \to 18 \to 94 \to 92 \to 86 \to 3 \to 50 \to 41 \to 66 \to 96 \to 0$
Heuristic	3	23	$0 \to 57 \to 81 \to 38 \to 28 \to 37 \to 61 \to 91 \to 25 \to 46 \to 5 \to 13 \to 1 \to 90 \to 15 \to 75 \to 12 \to 17 \to 88 \to 22 \to 80 \to 69 \to 10 \to 67 \to 0$
Heuristic	4	15	$0 \to 52 \to 97 \to 55 \to 62 \to 39 \to 45 \to 19 \to 31 \to 8 \to 36 \to 74 \to 78 \to 24 \to 76 \to 34 \to 0$
Simheuristic	1	19	$0 \to 30 \to 73 \to 16 \to 54 \to 58 \to 40 \to 64 \to 66 \to 41 \to 86 \to 92 \to 18 \to 94 \to 87 \to 7 \to 47 \to 3 \to 50 \to 96 \to 0$
Simheuristic	2	19	$0 \to 14 \to 10 \to 69 \to 80 \to 22 \to 88 \to 17 \to 12 \to 90 \to 1 \to 13 \to 5 \to 46 \to 25 \to 61 \to 26 \to 20 \to 67 \to 9 \to 0$
Simheuristic	3	15	$0 \to 57 \to 72 \to 77 \to 27 \to 95 \to 83 \to 37 \to 28 \to 38 \to 59 \to 42 \to 29 \to 81 \to 79 \to 23 \to 0$
Simheuristic	4	17	$0 \to 52 \to 85 \to 4 \to 62 \to 39 \to 45 \to 19 \to 31 \to 8 \to 68 \to 36 \to 24 \to 78 \to 74 \to 55 \to 97 \to 34 \to 0$

Deterministic validation on a small instance (Heuristic vs. exact MILP). To validate the quality of the proposed deterministic heuristic, we compare it against the exact MILP formulation on a reduced instance with 50 silos plus depot (51 nodes), with $m = 4$ available trucks and $T_{\max} = 15$. The MILP solver is initialized with the heuristic routing plan (warm start) and is run to check whether it can improve the incumbent solution.

Table 11 shows that the heuristic attains a total reward of 179 while visiting 11 silos in negligible time (and it leaves one truck unused, i.e., a null route $0 \to 0$). The MILP returns `Optimal` with the same objective value, hence it does not improve the heuristic solution and, instead, certifies its optimality on this instance. This experiment supports the claim that the constructive-first deterministic heuristic is already highly competitive, and that for this deliverable it is more beneficial to invest computational effort in stochastic evaluation (simheuristic layer) rather than introducing an additional deterministic metaheuristic phase as in the reference approach.

Table 12: Routes details for heuristic and MILP.

Algorithm	Truck	Route	Silos	Reward	Length	Slack
Heuristic	1	$0 \to 47 \to 3 \to 50 \to 41 \to 0$	4	69.00	12.28	2.72
Heuristic	2	$0 \to 23 \to 43 \to 34 \to 7 \to 0$	4	59.00	13.07	1.93
Heuristic	3	$0 \to 30 \to 9 \to 14 \to 0$	3	51.00	14.35	0.65
Heuristic	4	$0 \to 0$	0	0.00	0.00	15.00
MILP	1	$0 \to 43 \to 34 \to 7 \to 23 \to 0$	4	59.00	13.30	1.70
MILP	2	$0 \to 30 \to 0$	1	8.00	6.06	8.94
MILP	3	$0 \to 9 \to 14 \to 0$	2	43.00	12.00	3.00
MILP	4	$0 \to 41 \to 50 \to 3 \to 47 \to 0$	4	69.00	12.28	2.72

References

[1] J. Panadero, A. A. Juan, C. Bayliss, C. Currie, *Maximizing Reward from a Team of Surveillance Drones: a Simheuristic Approach to the Stochastic Team Orienteering Problem*, European Journal of Industrial Engineering 14.4 (2020): 485-516..

Table 11: Deterministic comparison between heuristic and MILP.

Algorithm	Status	Time (s)	Total reward	Silos	Trucks	Routes	Gap (%)
Heuristic	—	0.00	179.00	11	3	4	0.00
MILP	Optimal	1800.96	179.00	11	4	4	0.00